

BD® OMICS-One XT WTA Assay on the Hamilton™ Microlab™ NGS STAR Assay Ready Workstation

Streamline your single-cell gene expression
sample-to-answer workflow with automated
library preparation

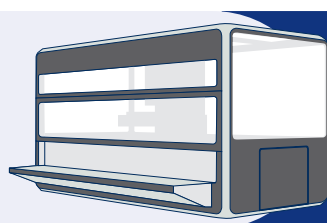
This innovative solution streamlines the process of single-cell whole transcriptome analysis, making it more efficient and accessible to more labs. With an optimized method for the BD® OMICS-One XT WTA Assay, the system is ready to use, ensuring high-quality results every time. The method's modular design allows easy integration with existing Hamilton™ Microlab™ NGS STAR Assay Ready Workstations (ARWs) and the BD Rhapsody™ System portfolio of instruments, reagents and bioinformatic tools, providing flexibility and scalability.

Your automation research partners, helping to streamline your work and drive greater insights



**BD® OMICS-One
XT WTA Assay Kit**
(Cat. No. 667246)

A cost-effective kit designed for automated library preparation, allowing comprehensive, high-throughput single-cell profiling, making large-scale studies more feasible



**Hamilton™ Microlab™
NGS STAR Assay
Ready Workstation
(ARW)**

The heart of the automation platform, providing efficient and reliable liquid handling capabilities with a biologically validated method for the BD® OMICS-One XT WTA Assay Kit



Scalable throughput

- Input more than 1 million cells
- Prepare up to 48 libraries at once



Standardized and reliable

Benefit from an optimized method that ensures consistent, quality libraries



Modular flexibility

Easily integrate the BD® OMICS-One XT WTA Assay method with your existing Hamilton™ Microlab™ NGS STAR ARW



Flexible and modular workflow for stress-free, walkaway processing with multiple safe stopping points

Random Priming and Extension (RPE) and RPE PCR



WTA Index PCR

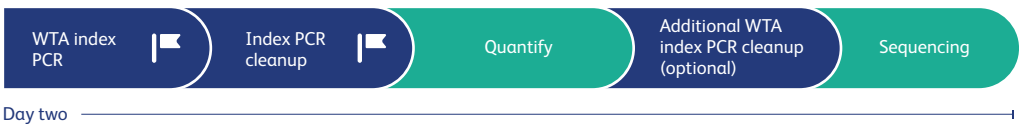
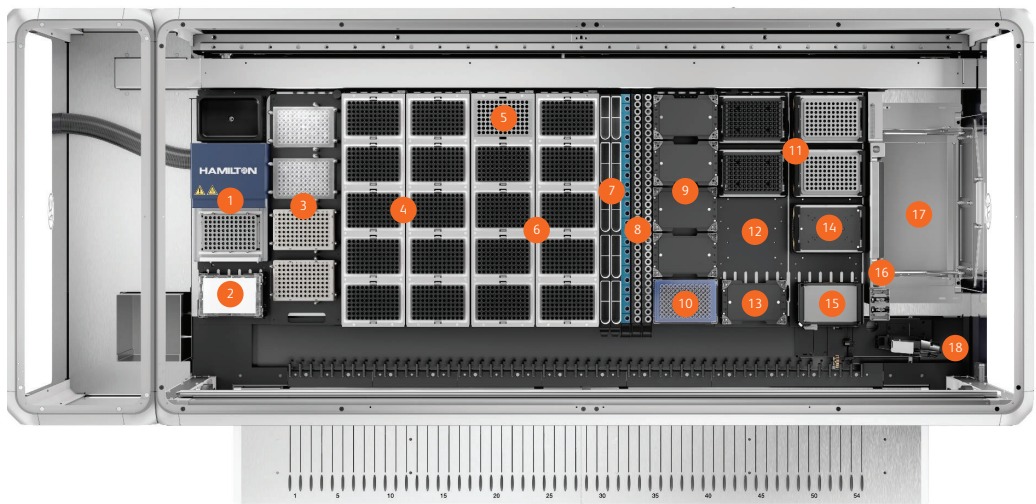


Figure 1. The BD® OMICS-One XT WTA Assay method on the Hamilton® Microlab® NGS STAR ARW is designed for maximum flexibility and modularity. The method allows a user multiple suitable safe stopping points to avoid extended hours in the lab, breaking up the workflow to fit busy schedules, while ensuring sample integrity. To generate sequencing-ready libraries in under two typical eight-hour shifts, it is recommended to only process up to 24 libraries at a time.

System details and configuration



- | | | |
|-----------------------------------|----------------------------|----------------------------|
| 1 96-Well On-Deck Thermal Cycler* | 7 60-mL Reagent Troughs | 13 Cold Plate (Air Cooled) |
| 2 Stacked Plate Lids | 8 Microtubes | 14 Heater Shaker |
| 3 Stacked Plates | 9 Plate Positions | 15 Cold Plate (Air Cooled) |
| 4 50-µL Conductive Tips | 10 Plate Magnet | 16 CO-RE™ Paddles |
| 5 Tip Adapter* | 11 Heater Shakers | 17 Channel Waste |
| 6 300-µL Conductive Tips | 12 Cold Plate (Air Cooled) | 18 Autoloader |

*On-Deck Thermal Cycler and 96-Multi-Probe Head are optional

Figure 2. The Hamilton® Microlab® NGS STAR ARW, designed and optimized for NGS library preparation. The system is configured with eight independent 1000-µL pipetting channels and comes with an autoloader barcode reader and plate transfer capabilities using the internal plate gripper (iSWAP) and Hamilton's CO-RE™ Paddles. It is designed and optimized to run NGS library preparation methods with devices and accessories needed for sample tracking, reagent handling, heating and cooling steps and magnetic bead-based cleanups. The system can also accommodate optional on-deck thermal cycling and is pre-configured for a 96 multi-probe pipetting head addition.

Validated with strict QC measures and reproducible, high-quality results

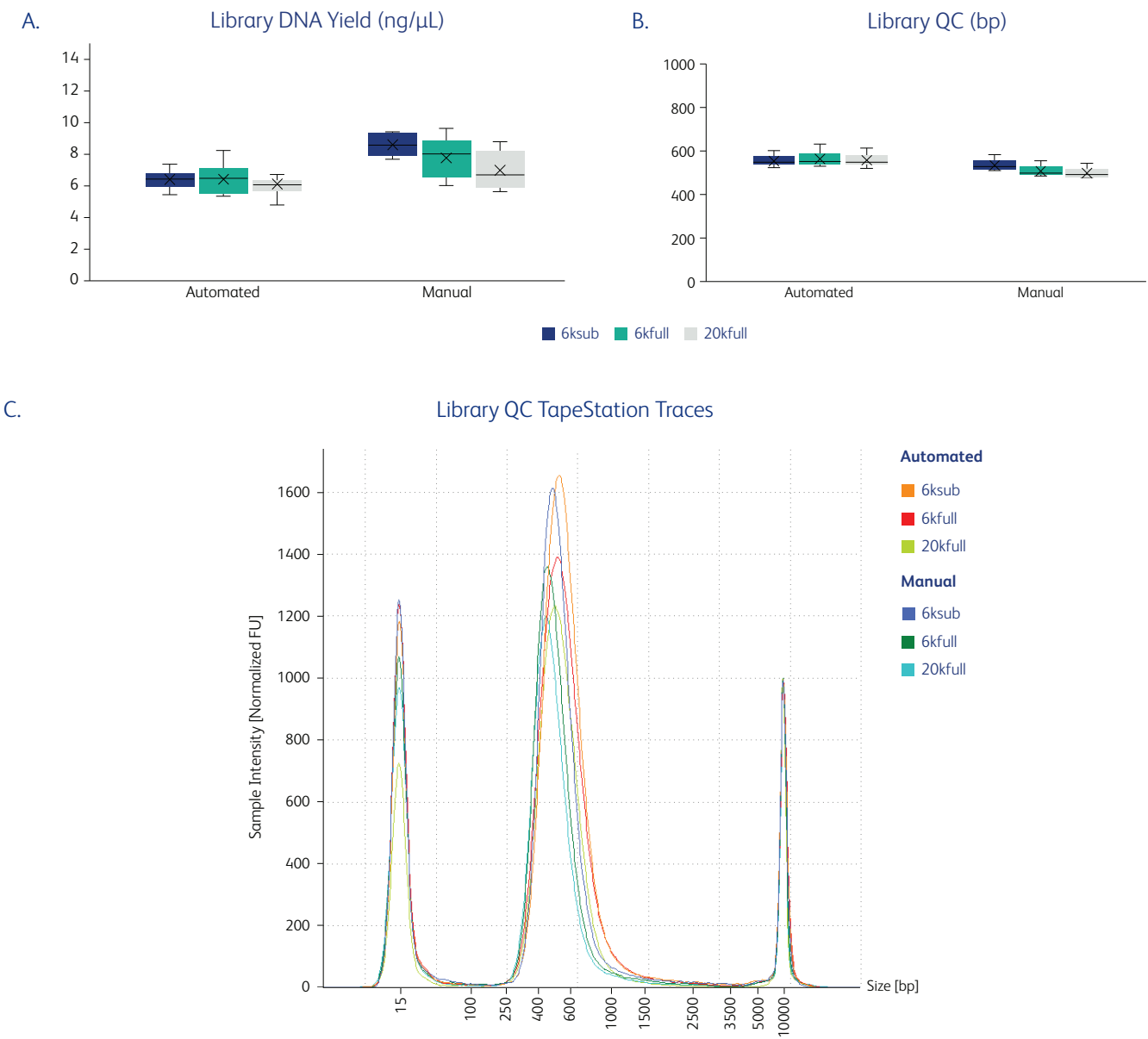


Figure 3. Strict quality control measures in place, ensuring optimal concentrations of libraries and correct distribution of fragment sizes. Three runs of 18 libraries (six each of 6K subsampled beads, 6K full sampled beads and 20K full sampled beads from the same pool of cellular transcripts on the BD Rhapsody™ Enhanced Cell Capture Beads) were prepared on the Hamilton™ Microlab™ NGS STAR ARW along with five manual bench controls of each condition. **A)** Final quality control of indexed libraries shows automated libraries were comparable to manual bench controls for Qubit concentration (ng/μL). **B)** Final quality control by TapeStation analysis show similar average bp size distribution between 200–1,000bp. **C)** Final indexed libraries were diluted to 1 ng/μL prior to running on the D5000 High Sensitivity TapeStation System. All cell inputs, automated and manual, have equivalent base pair size distribution.

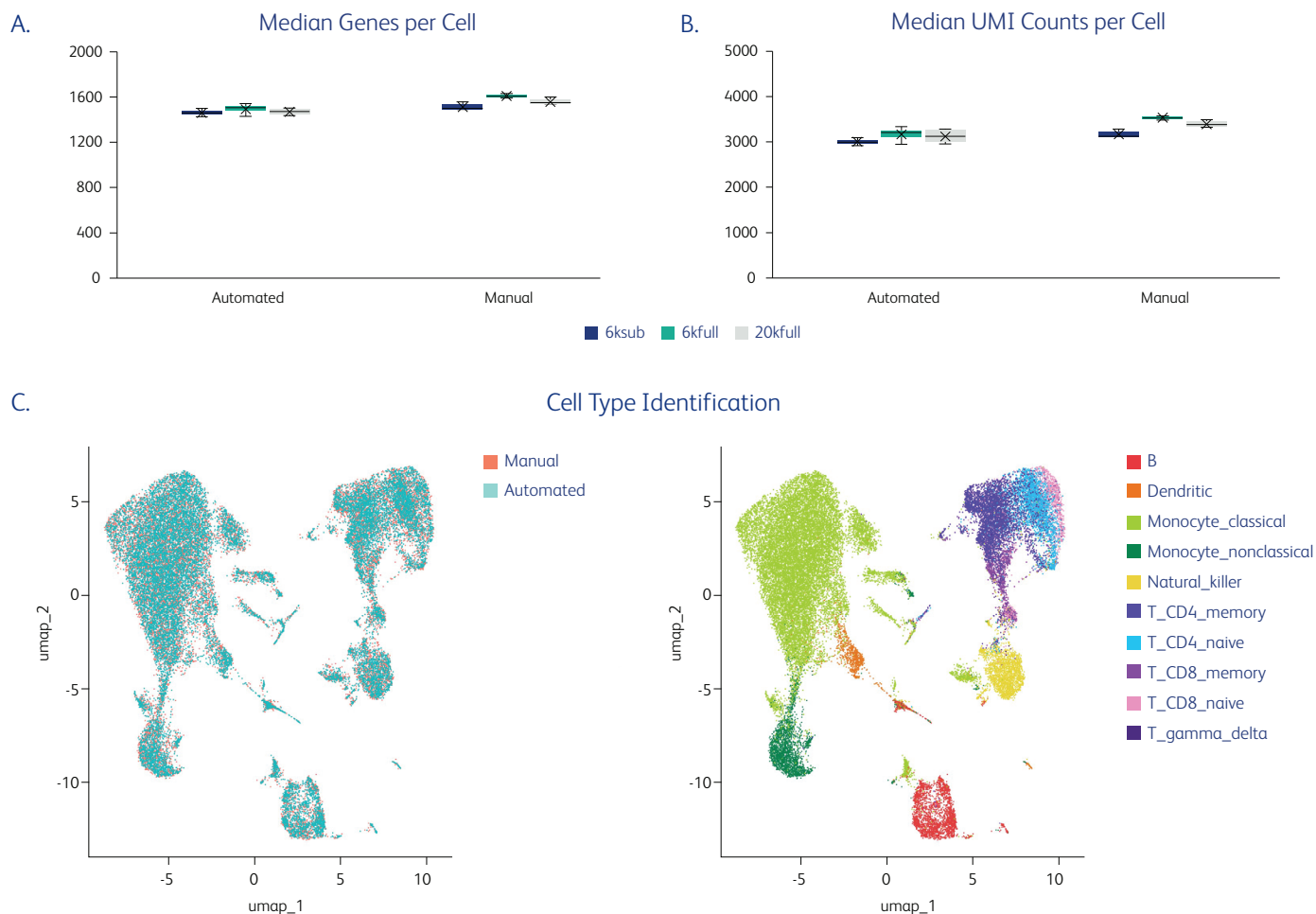


Figure 4. Optimized and biologically validated methods ensure high-quality results every time. Three runs of 18 libraries (six each of 6K subsampled beads, 6K full sampled beads and 20K full sampled beads from the same pool of cellular transcripts on the BD Rhapsody[™] Enhanced Cell Capture Beads) were prepared on the Hamilton[™] Microlab[™] NGS STAR ARW along with five manual bench controls of each condition. Shown are sequenced libraries—15 libraries total for 6K sub and 6K full each and nine libraries total for 20K full sampled beads. After sequencing, all automated and manual libraries were down sampled to between 7,600–8,000 raw reads per cell. **A)** Median genes per cell generated from libraries prepped on the Hamilton[™] Microlab[™] NGS STAR ARW are comparable to manually prepped libraries. **B)** Median UMI counts per cell generated on the automated system are like those libraries generated manually. **C)** UMAP of 20K cell input show no batch effects between automated and manual libraries and cell types are successfully identified. Similar results (not shown) were observed for both 6K sub and 6K full sampled beads.

Additional resources

For any additional questions, contact scomix@bdscomix.bd.com or contact your local BD sales representative.

Check out our product page to learn more



For Research Use Only. Not for use in diagnostic or therapeutic procedures.

BD Life Sciences, Milpitas, CA 95035, U.S.

bdbiosciences.com

